

Python Tip: `zip`

Need to loop over two lists at the same time?

Use `zip()` to pair related values together.

Without `zip`

You can use indexes to read from two lists, but it makes the loop harder to read.

```
names = ["Ada", "Grace", "Katherine"]  
scores = [98, 95, 99]  
  
for i in range(len(names)):  
    print(names[i], scores[i])
```

```
Ada 98  
Grace 95  
Katherine 99
```

With `zip`

`zip()` pairs the first item from each list, then the second, then the third.

```
names = ["Ada", "Grace", "Katherine"]
scores = [98, 95, 99]

for name, score in zip(names, scores):
    print(name, score)
```

```
Ada 98
Grace 95
Katherine 99
```

Cleaner loop. Same result.

More Than Two Lists

You can zip three or more lists when each item has multiple related values.

```
names = ["Ada", "Grace", "Katherine"]
scores = [98, 95, 99]
grades = ["A", "A", "A"]

for name, score, grade in zip(names, scores, grades):
    print(f"{name}: {score} ({grade})")
```

```
Ada: 98 (A)
Grace: 95 (A)
Katherine: 99 (A)
```


Watch the Lengths

`zip()` stops when the shortest list runs out of items.

```
names = ["Ada", "Grace", "Katherine"]  
scores = [98, 95]  
  
for name, score in zip(names, scores):  
    print(name, score)
```

```
Ada 98  
Grace 95
```

`Katherine` is skipped because there is no matching score.

Combine with `enumerate`

Use `enumerate()` with `zip()` when you also need a counter.

```
names = ["Ada", "Grace", "Katherine"]  
scores = [98, 95, 99]  
  
for rank, (name, score) in enumerate(zip(names, scores)):  
    print(f"{rank}. {name}: {score}")
```

```
1. Ada: 98  
2. Grace: 95  
3. Katherine: 99
```

Wrap-Up

Now you can use `zip()` to loop over related lists together.

Follow me for more tips.

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